

Trainings ▾

General Information

ReCon® has implemented a new stamping code for all K and QSK cylinder heads.

The stamp is a diamond shape located between the short port exhaust valve and long port intake valve on the rocker lever side of the cylinder head. The marking (or lack of a marking) inside the diamond shape indicates the size of the injector seal.

- An empty diamond shape indicates that the ReCon® cylinder head is equipped with a standard size injector seal.
- One mark inside the diamond shape indicates the cylinder head is equipped with a size 0.010 injector seal.
- Two marks inside the diamond shape indicate that the cylinder head is equipped with a size 0.020 injector seal.
- Three marks inside the diamond shape indicate that the cylinder head is equipped with a size 0.030 injector seal.

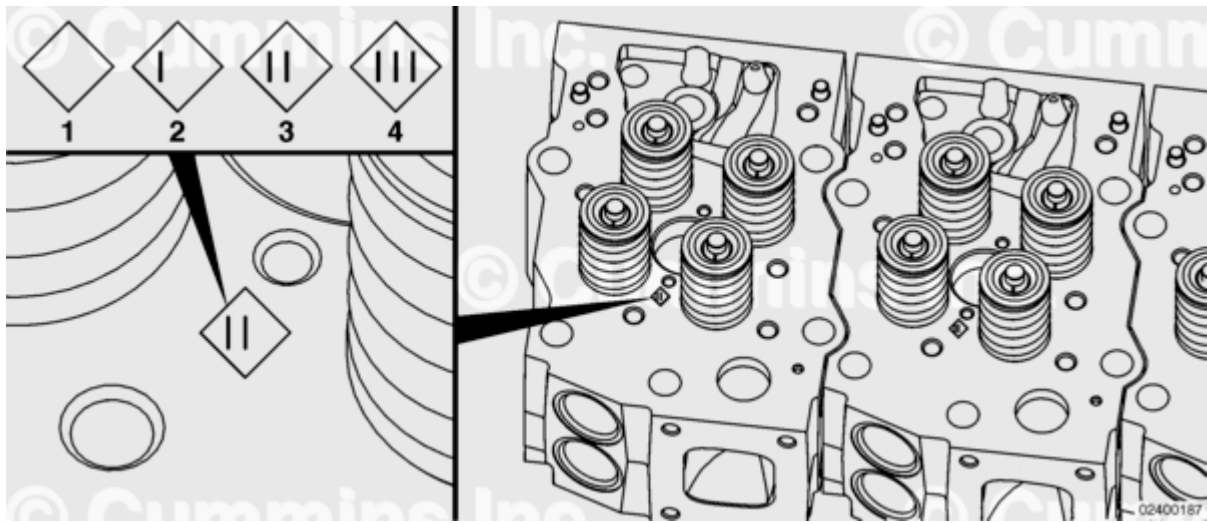
K and QSK cylinder heads use a different injector seal. Reference the tables below.

K Cylinder Head Seal Part Numbers

Injector Seal Part Number	Injector Seal Size
207244	Standard
3001658	0.010 oversize
3001659	0.020 oversize
3001660	0.030 oversize

QSK (with Mechanically Actuated Injectors) Cylinder Head Seal Part Numbers

Injector Seal Part Number	Injector Seal Size
3867687	Standard
3347933	0.010 oversize
3347934	0.020 oversize
3347935	0.030 oversize



Injector Seal Markings Location

Injector Seal Markings

1. Standard
2. 0.010 oversize
3. 0.020 oversize
4. 0.030 oversize.

Preparatory Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



ck800wa

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

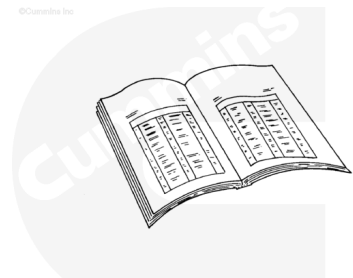
- Disconnect the batteries to prevent accidental engine starting. Refer to the OEM service manual.
- Disconnect the air supply to the air starter.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)
- Remove the turbocharger. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-033-tr.html>)
- Remove the exhaust manifold. Refer to Procedure 011-007 in Section 11. (</qs3/pubsys2/xml/en/procedures/20/20-011-007-tr.html>)
- Remove the fuel tubes. Refer to Procedure 006-024 in Section 6. (</qs3/pubsys2/xml/en/procedures/20/20-006-024-tr.html>)
- Remove the aftercooler. Refer to Procedure 010-002 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-010-002-tr.html>)
- Remove the fuel manifolds. Refer to Procedure 006-022 in Section 6. (</qs3/pubsys2/xml/en/procedures/20/20-006-022-tr.html>)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html>)
- Remove the rocker levers. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-009-tr.html>)
- Remove the valve push rods, injector push rod, and crossheads. Refer to Procedure 004-014 in Section 4. (</qs3/pubsys2/xml/en/procedures/20/20-004-014-tr.html>)
- Remove the coolant thermostat housing support. Refer to Procedure 008-015 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-015-tr.html>)
- Remove the gear cover clamping plate. Refer to Procedure 001-029 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-029-tr.html>)

- Remove the rocker lever housing. Refer to Procedure 003-013 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-013-tr.html>)
- Remove the injectors. Refer to Procedure 006-026 in Section 6. (</qs3/pubsys2/xml/en/procedures/20/20-006-026-tr.html>)

Marine Applications

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



ck800wa

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift component or assembly.

⚠ CAUTION ⚠

Use caution when draining coolant that the coolant is not spilled or drained into the bilge area. Do not pump coolant overboard. If the coolant is not reused, it must be disposed of in accordance with local environmental regulations.

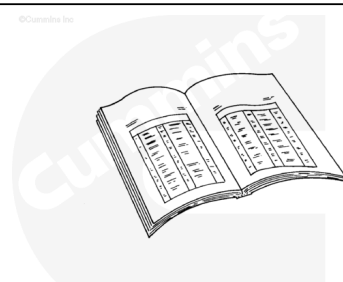
- Disconnect the batteries to prevent accidental engine starting. Refer to the OEM service manual.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)
- Remove the crankcase breathers. Refer to Procedure 003-001 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-001-tr.html>)
- Remove the exhaust manifold heat shields. Refer to Procedure 011-015 in Section 11. (</qs3/pubsys2/xml/en/procedures/20/20-011-015-tr.html>)
- Remove the turbocharger heat shields. Refer to Procedure 010-076 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-076-tr.html>)
- Remove the aftercooler heat shields. Refer to Procedure 010-129 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-129.html>)
- Remove the turbocharger. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-033-tr.html>)
- Remove the exhaust manifold. Refer to Procedure 011-007 in Section 11. (</qs3/pubsys2/xml/en/procedures/20/20-011-007-tr.html>)
- Use the Troubleshooting and Repair Manual, Electronic Control System, QSK19 CM850, Modular Common Rail System Series Engines, Bulletin 4021493, for information to disconnect the wiring harness in the areas that restrict access to the cylinder head. Refer to Procedure 019-043 in Section 19. (</qs3/pubsys2/xml/en/procedures/13/13-019-043.html>)
- Remove the aftercooler. Refer to Procedure 010-002 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-002-tr.html>)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html>)
- Remove the rocker levers. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-009-tr.html>)

- Remove the push rods and crossheads. Refer to Procedure 004-014 in Section 4. (</qs3/pubsys2/xml/en/procedures/20/20-004-014-tr.html>)
- Remove the rocker lever housing. Refer to Procedure 003-013 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-013-tr.html>)
- Remove the injectors. Refer to Procedure 006-026 in Section 6. (</qs3/pubsys2/xml/en/procedures/20/20-006-026-tr.html>)

with Electronically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



ck800wa

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Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries to prevent accidental engine starting. Refer to the OEM service manual.
- Disconnect the air supply to the air starter.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)
- Remove the turbocharger. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-033-tr.html>)

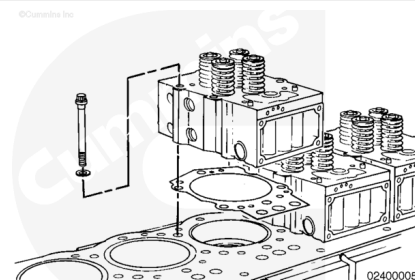
- Remove the exhaust manifold (dry), if installed. Refer to Procedure 011-007 in Section 11.
(/qs3/pubsys2/xml/en/procedures/20/20-011-007-tr.html)
- Remove the exhaust manifold, (wet), if installed. Refer to Procedure 011-008 in Section 11.
(/qs3/pubsys2/xml/en/procedures/20/20-011-008-tr.html)
- Remove the high-pressure fuel lines to injectors. Refer to Procedure 006-051 in Section 6.
(/qs3/pubsys2/xml/en/procedures/20/20-006-051-tr.html)
- Remove the fuel drain lines. Refer to Procedure 006-013 in Section 6. (/qs3/pubsys2/xml/en/procedures/20/20-006-013-tr.html)
- Remove the intake manifold. Refer to Procedure 010-023 in Section 10. (/qs3/pubsys2/xml/en/procedures/20/20-010-023-tr.html)
- Remove the coolant thermostat housing. Refer to Procedure 008-015 in Section 8.
(/qs3/pubsys2/xml/en/procedures/20/20-008-015-tr.html)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html)
- Remove the rocker levers. Refer to Procedure 003-009 in Section 3. (/qs3/pubsys2/xml/en/procedures/20/20-003-009-tr.html)
- Remove the valve push rods, injector push rod, and crossheads. Refer to Procedure 004-014 in Section 4.
(/qs3/pubsys2/xml/en/procedures/20/20-004-014-tr.html)
- Remove the coolant thermostat housing support. Refer to Procedure 008-015 in Section 8.
(/qs3/pubsys2/xml/en/procedures/20/20-008-015-tr.html)
- Remove the gear cover clamping plate. Refer to Procedure 001-029 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-029-tr.html)
- Remove the rocker lever housing. Refer to Procedure 003-013 in Section 3.
(/qs3/pubsys2/xml/en/procedures/20/20-003-013-tr.html)

- Remove the injectors. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/20/20-006-026-tr.html)

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift component or assembly.



Remove the six cylinder head capscrews.

Remove the cylinder head and gasket.

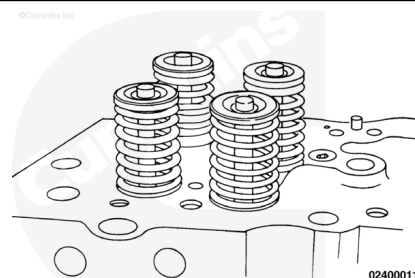
Record the cylinder head gasket part number to determine if the gasket is a standard or oversize thickness.

Discard the gasket.

Clean

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Clean the cylinder heads with solvent, Part Number 3924421, or equivalent.

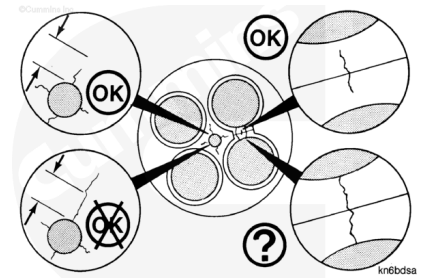
Check for broken springs and other damage.

Inspect for Reuse

with Mechanically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Check for cracks on the combustion surface.

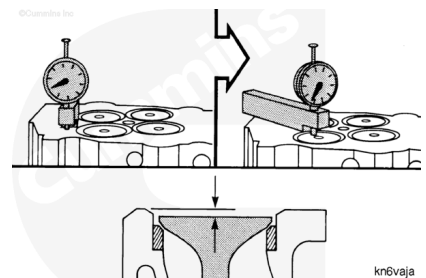
If a crack around the injector bore exceeds the maximum length, the cylinder head **must** be replaced.

Allowable Crack Length		
mm		in
6	MAX	0.25

Both ends of a crack between the valves **must** be visible. If one end of a crack extends into the valve seat bore (behind the valve seat), the condition of the cylinder is questionable. Remove the valve and valve seat to make sure the cylinder head is reusable.

Measure the valve head depth in the cylinder head with depth gauge, Part Number 3164438, or equivalent.

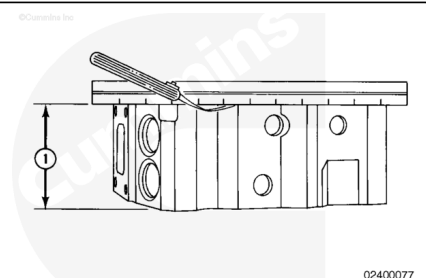
Allowable Valve Depth from the Combustion Face		
mm		in
0.00	MIN	0.000
0.51	MAX	0.020



If the valve head depth is **not** within specifications, the cylinder head **must** be reconditioned.

Check the flatness of the cylinder head with a straightedge and feeler gauge.

The cylinder head **must** be resurfaced or replaced if a feeler gauge larger than 0.08 mm [0.003 in] will fit between the straightedge and the cylinder head.



The cylinder head can be resurfaced **only** on mechanically actuated injectors, if the head thickness measurement is within specifications.

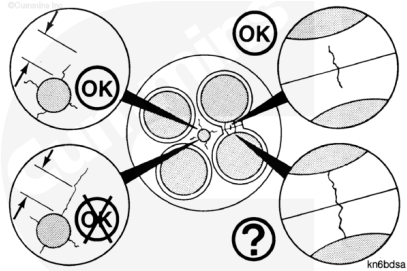
QSK19 Mechanically Actuated Injector Cylinder Head Thickness

	mm		in
Used Cylinder Head Service Limit	150.11	MIN	5.910
New Cylinder Head	150.88	MIN	5.940
	151.12	MAX	5.950

with Electronically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Check for cracks on the combustion surface.

If a crack around the injector bore exceeds the maximum length, the cylinder head **must** be replaced.

Allowable Crack Length

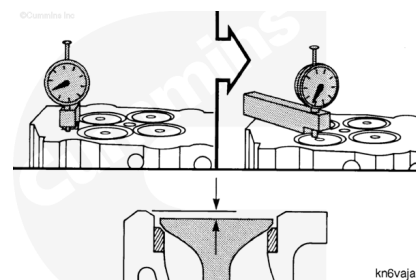
mm		in
6	MAX	0.25

Both ends of a crack between the valves **must** be visible. If one end of a crack extends into the valve seat bore (behind the valve seat), the condition of the cylinder is questionable. Remove the valve and valve seat to make sure the cylinder head is reusable.

Measure the valve head depth in the cylinder head with depth gauge, Part Number 3164438, or equivalent.

Allowable Valve Depth from the Combustion Face

mm		in
0.00	MIN	0.000
0.51	MAX	0.020

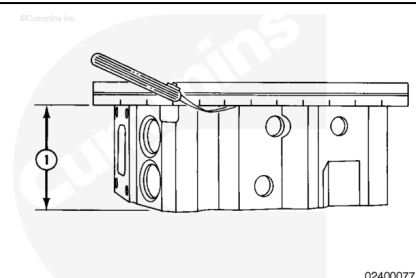


If the valve head depth is **not** within specifications, the cylinder head **must** be reconditioned.

Check the flatness of the cylinder head with a straightedge and feeler gauge.

The cylinder head **must** be replaced if a feeler gauge larger than 0.08 mm [0.003 in] will fit between the straightedge and the cylinder head.

The cylinder head can be resurfaced **only** on mechanically actuated injectors, if the head thickness measurement is within specifications.



QSK19 Electronically Actuated Injector Cylinder Head Thickness

	mm		in
New Cylinder Head	120.52	MIN	4.745
	120.78	MAX	4.755

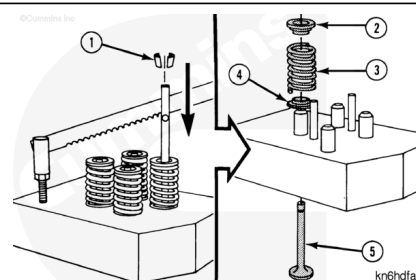
Note : Cylinder heads with electronically actuated injectors can **not** be resurfaced, as the injector protrusion can **not** be adjusted.

Disassemble

with Mechanically Actuated Injector

⚠ WARNING ⚠

The valve springs are under compression. Use care when using the valve spring compressor. Personal injury can result if the tool slips from your hands.



Use one of the valve spring compressors listed below to remove the valve springs.

- Valve spring compressor, Part Number 3163606, or equivalent
- Valve spring compressor stand, Part Number ST-1022, or equivalent
- Valve spring compressor plate, Part Number 3163177, or equivalent, can be used with valve spring compressor, Part Number 3163606, or equivalent, and valve spring compressor stand, Part Number ST-1022, or equivalent, to remove four springs at a time
- Air operated valve spring compressor, Part Number 3375960.

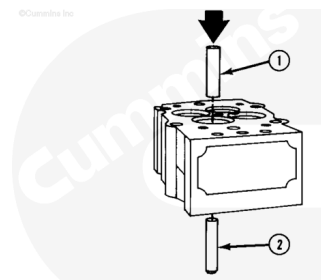
The intake and exhaust valves are different.

Mark the valves for location prior to removal to aid in assembly.

Remove the listed parts:

1. Valve collet
2. Valve spring retainer
3. Valve spring
4. Valve spring rotator/or guide
5. Valve

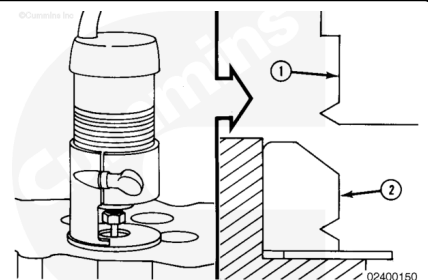
If the valve guide is replaced, the valve seat insert for the valve guide **must** be measured for runout. It is likely the seat **must** also be machined or replaced.



Note : Only replace the valve seat if it is not within specifications.

If necessary, a groove can be machined in the valve seat insert (2) to allow the valve seat extractor (1) to be used.

Machine a groove into the valve seat insert, as close to the bottom of the bore as possible, with valve seat grooving kit, Part Number 3376405, or equivalent.

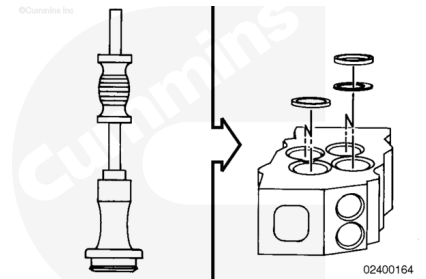


Remove the valve seat with the appropriate valve seat extractor listed below and slide hammer, Part Number 3376799:

- Exhaust seat extractor, Part Number ST-1323-1
- Intake seat extractor, Part Number 3376799.

Note : On mechanically actuated injector engines, the intake ports contain anti-swirl plates.

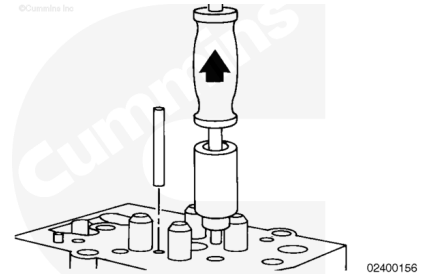
Remove and discard the anti-swirl plate, if used.



Note : This step is only for the mechanically actuated injector engine.

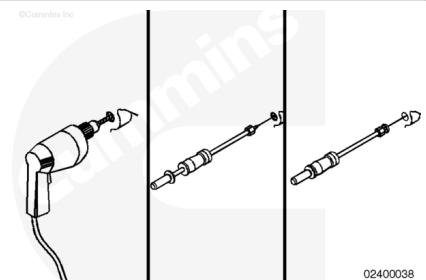
Note : Only remove the crosshead guide if it is bent or is not within specifications.

Crosshead guides that are straight and are installed to correct height do **not** have to be removed to install stemless crossheads.



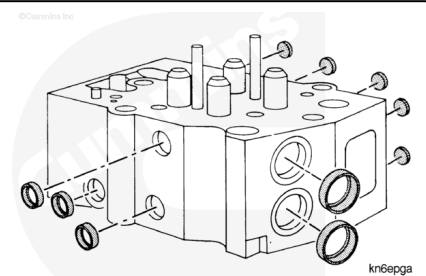
Note : Do not remove the cup plugs if a pressure test is to be performed.

Use a drill, sheet metal screw, and a slide hammer from the light duty puller kit, Part Number 3375784, or equivalent, to remove the cup plugs.



Cup plugs **must** be removed from the head casting for cleaning purposes.

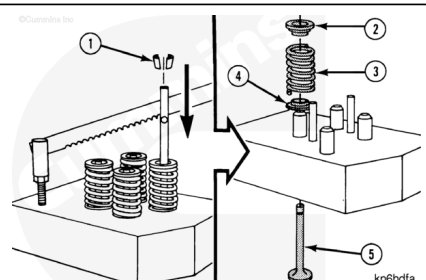
Remove and discard the ten cup plugs.



with Electronically Actuated Injector

⚠ WARNING ⚠

The valve springs are under compression. Use care when using the valve spring compressor. Personal injury can result if the tool slips from your hands.



Use one of the valve spring compressors listed below to remove the valve springs.

- Valve spring compressor, Part Number 3163606, or equivalent
- Valve spring compressor stand, Part Number ST-1022, or equivalent
- Valve spring compressor plate, Part Number 3163177, or equivalent, can be used with valve spring compressor, Part Number 3163606, or equivalent, and valve spring compressor stand, Part Number ST-1022, or equivalent, to remove four springs at time
- Air operated valve spring compressor, Part Number 3375960.

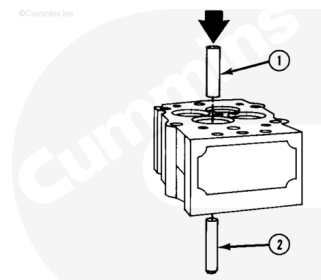
The intake and exhaust valves are different.

Mark the valves for location prior to removal to aid in assembly.

Remove the listed parts:

1. Valve collet
2. Valve spring retainer
3. Valve spring
4. Valve spring rotator/or guide
5. Valve
6. Valve stem seal (**not** shown) (used **only** on the electronically actuated injector engine).

If the valve guide is replaced, the valve seat insert for the valve guide **must** be measured for runout. It is likely the seat **must** also be machined or replaced.

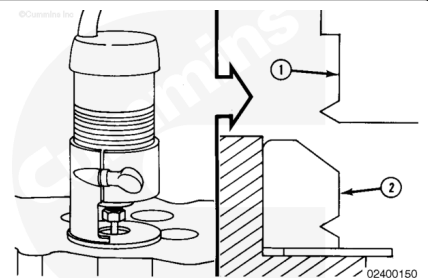


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Note : Only replace the valve seat if it is not within specifications.

If necessary, a groove can be machined in the valve seat insert (2) to allow the valve seat extractor (1) to be used.

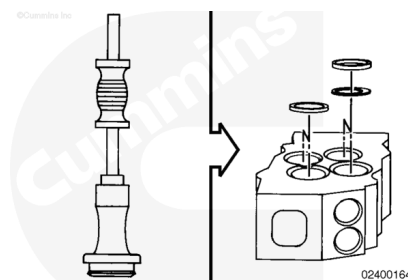
Machine a groove into the valve seat insert, as close to the bottom of the bore as possible, with valve seat grooving kit, Part Number 3376405, or equivalent.



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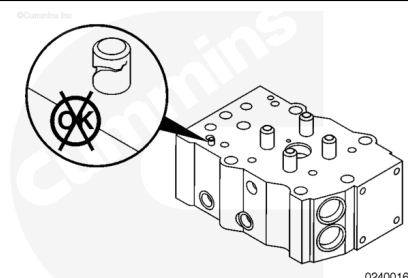
Remove the valve seat with the appropriate valve seat extractor listed below and slide hammer, Part Number 3376799:

- Exhaust seat extractor, Part Number ST-1323-1
- Intake seat extractor, Part Number 3376799.



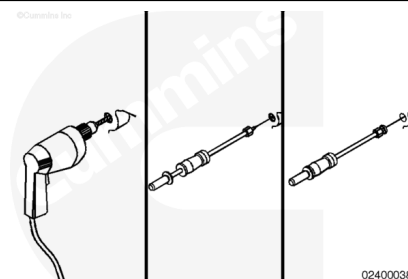
Note : This step is **only** for the electronically actuated injector engine.

Remove the dowel pins **only** if they are damaged or **not** installed to the correct height.



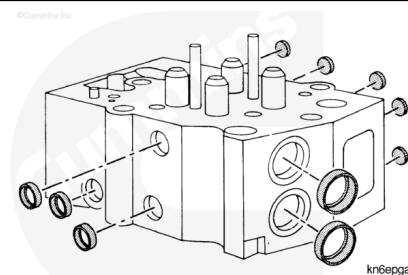
Note : Do not remove the cup plugs if a pressure test is to be performed.

Use a drill, sheet metal screw, and a slide hammer from the light duty puller kit, Part Number 3375784, or equivalent, to remove the cup plugs.



Cup plugs **must** be removed from the head casting for cleaning purposes.

Remove and discard the ten cup plugs.

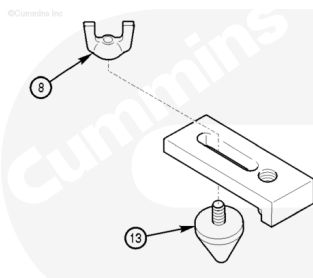


Pressure Test

Preferred Method

Use the cylinder head leak test kit, Part Number 3164341, or equivalent, with the pressure regulator valve kit, Part Number 3164231, or equivalent, to pressure test the cylinder head.

Assemble the wing nut (8) to the clamp and plug (13).



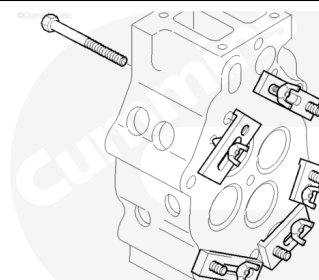
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Install the clamp and plug assembly into the cylinder head to block the coolant passages.

Insert the appropriate capscrew through the valve guide side of the cylinder head and thread the capscrew into the clamp.

Tighten the capscrew.

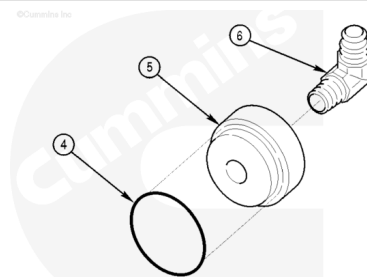
Torque Value: 20 n•m [177 in-lb]



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Install the o-ring (4) into the o-ring groove of the test adapter (5).

Thread the adapter elbow (6) into the adapter (5).



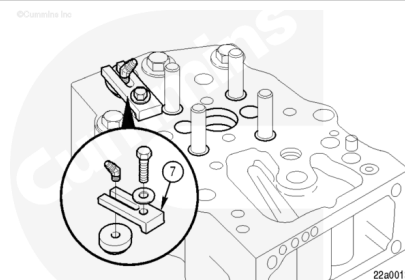
22A00133

Place the test adapter assembly into the cylinder head.

Install the clamp (7) over the test adapter assembly and secure it with the appropriate capscrew and washer.

Tighten the capscrew.

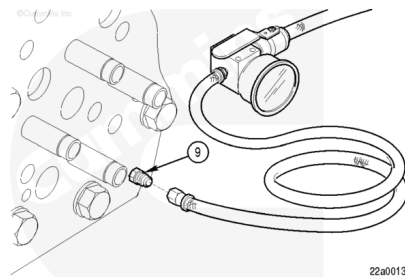
Torque Value: 54 n•m [40 ft-lb]



22a00134

Connect the pressure regulator valve kit to the pipe nipple.

Connect the pressure regulator valve kit to shop air and adjust the air pressure.

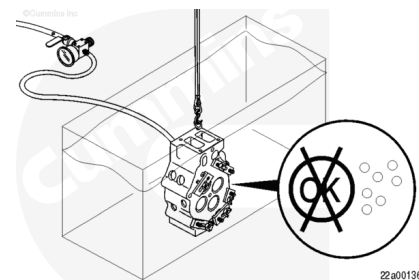


22a00135

Measurements

	kpa	psi
Pressure Regulator Valve	345	50

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Attach an eye bolt to the cylinder head.

Use a suitable lifting device to lift the cylinder head.

Immerse the cylinder head into a tank of water.

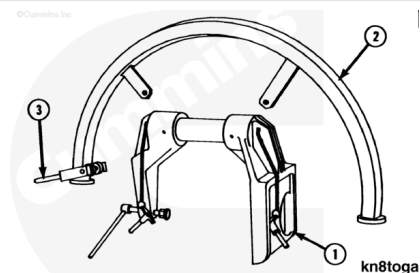
Check for bubbles. Bubbles indicate an air leak.

If the cylinder head leaks, it **must** be repaired or replaced.

Optional Method

One method to pressure test the cylinder head is to use the hydrostatic tester, Part Number ST-1012, or equivalent, with the water test adapter plate, Part Number 3375070.

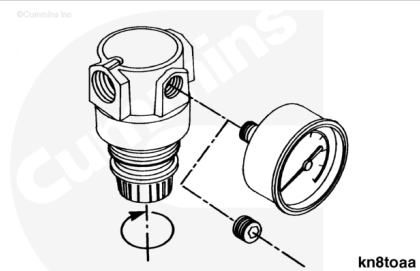
The following steps outline this method.



Remove one of the two plugs from the pressure regulator.

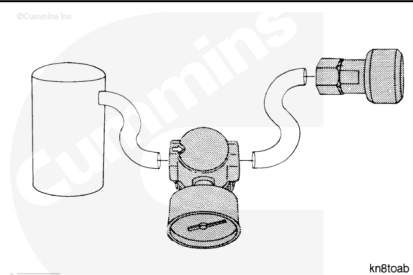
Install the pressure gauge in the regulator.

Turn the adjusting knob on the regulator **counterclockwise**, as far as it will turn.



Install the pressure regulator between the air supply and the quick disconnect fitting.

The arrow on the top of the pressure regulator **must** point in the direction of the air flow (toward the quick disconnect fitting).

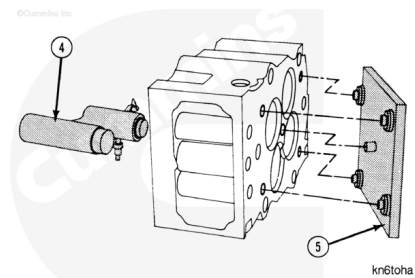


With the head positioned as shown in the illustration, assemble the adapter plates.

The guide pins on the lower plate (5), Part Number 3375070, or equivalent, **must** fit into the water passages.

The o-ring on the upper plate (4) creates a seal on the upper water passage.

The other end of the plate fits into the injector bore.

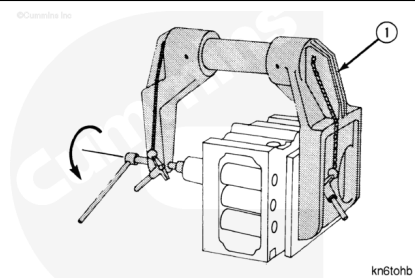


Place the clamping assembly (1) over the head and adapters.

The guide pins on the clamp **must** fit into the holes in the adapter plate.

Tighten the clamp on the cylinder head.

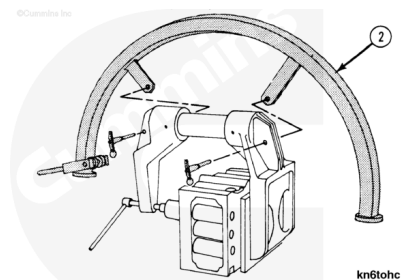
Connect the regulated air supply hose to the upper adapter fitting.



Attach the lifting arm (2) to the clamp assembly with the lock pins.

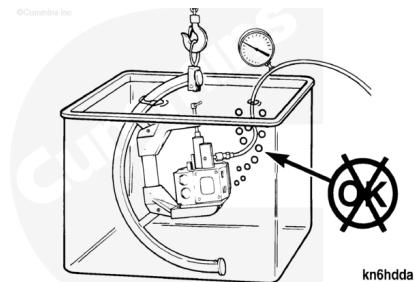
One mounting location on each piece is color-coded red.

Match the red locations.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift component or assembly.



Adjust the air pressure.

Air Pressure		
kpa		psi
552	MAX	80

Lower the cylinder head into the tank of water.

Bubbles will indicate a leak.

If the cylinder head leaks, it **must** be repaired or replaced.

Clean and Inspect for Reuse

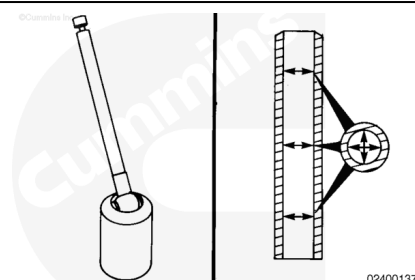
with Mechanically Actuated Injector

Check the valve guide for chips and cracks.

Measure the inside diameter of the guide with a ball gauge or a dial bore indicator.

Valve Guide Inside Diameter (Installed)		
mm		in
10.561	MIN	0.4158
10.589	MAX	0.4169

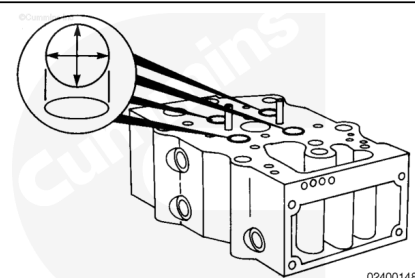
The valve guide can be reusable if the inside diameter of the first 13 mm [0.50 in] from the top or bottom of the valve guide is over specification. However, if an area 13 mm [0.50 in] or greater from the top or bottom is out of tolerance, the valve guide **must** be replaced.



02400137

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



02400148

Check the valve guide for chips and cracks.

Use the drop gauge, Part Number 5298508, to measure the valve guide per the assembly procedure below.

If the valve guide is out of tolerance, the valve guide **must** be replaced.

Clean the part with solvent, Part Number 3824421, or equivalent.

Measure the valve guide bore inside diameter.

Valve Guide Bore Inside Diameter		
mm		in
21.438	MIN	0.844
21.463	MAX	0.845

If the valve guide bore is **not** within specifications, ream the bore for an oversize valve guide.

The oversize valve guides are oversize on the outside diameter **only**. The bore in the valve guide is **not** oversize.

Valve guides are available in two oversizes:

- 0.25 mm [0.010 in]
- 0.38 mm [0.015 in].

Ream the valve guide bore to maintain a press fit within 0.028 to 0.067 mm [0.001 to 0.003 in] between the valve guide and the valve guide bore.

Clean the valve seat insert bore.

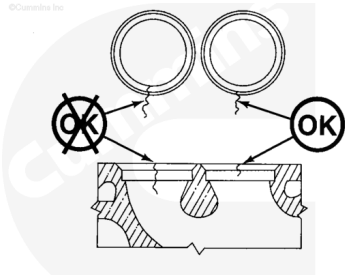
Check the length of any cracks extending into the valve insert bore.

The cylinder head **must** be replaced if a crack extends into the bottom of the bore.

Sometimes it is possible to remove the crack by machining the cylinder head to use an oversize valve seat insert.

Use valve insert counterbore cutter kit, Part Number ST-257, or equivalent, along with appropriate cutter listed below:

- ST-1310-1 (0.010 oversize)
- ST-1310-2 (0.020 oversize)
- ST-1310-3 (0.030 oversize)
- ST-1310-4 (0.040 oversize).

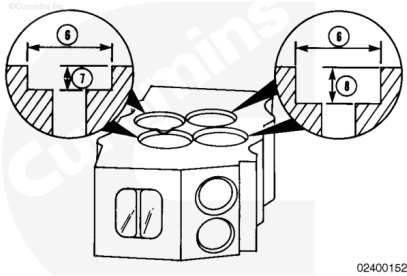


02400161

Measure the inside diameter and the depth of the valve seat insert bore.

Valve Seat Insert Bore Inside Diameter (6)			
	mm		in
Exhaust and Intake	60.37	MIN	2.377
	60.40	MAX	2.378

Valve Seat Insert Bore Depth (7) and (8)			
	mm		in
Exhaust and Intake	12.50	MIN	0.492
	12.62	MAX	0.497



02400152

If the valve guide seat insert inside diameter is **not** within specifications, use an oversize valve seat.

Oversize valve seat inserts are available in the sizes listed in the table below. Machine the valve seat insert bore in the cylinder head to maintain a press fit within 0.064 to 0.114 mm [0.002 to 0.005 in] between the valve seat insert and valve seat insert bore.

Outside Diameter Oversize	Depth (Thickness) of Oversize
0.25 mm [0.010 in]	Standard
0.51 mm [0.020 in]	0.13 mm [0.005 in]
0.76 mm [0.030 in]	0.25 mm [0.010 in]
1.02 mm [0.040 in]	0.38 mm [0.015 in]

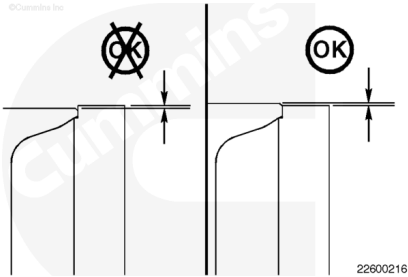
The intake and exhaust valve face and seating surfaces **must** be cleaned and free of carbon deposits before the inspection is performed.

Use Scotch-Brite™ 7448 abrasive hand pad, Part Number 3823258, or equivalent, to clean the valve.

Use the valve head checking tool, Part number 3164984, to measure the head thickness of the valve.

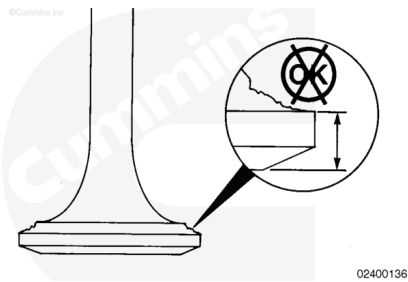
Place the valve into the valve head thickness gauge.

Measure the valve height. If the valve is flush or above the gauge surface, the valve can be reused. If the valve is below the gauging surface, the valve can **not** be reused.



⚠ CAUTION ⚠

This type of a check is not as accurate as the checking tool. It can result in valves that are not within minimum specification being reused. Valves that are not within minimum specification can fail, causing severe progressive damage to the power cylinder.



If a valve checking tool is **not** available, place the valve on a flat surface and check the height of the outside diameter.

Valve Head Thickness (at the outside diameter)

	mm		in
	3.94	MIN	0.155
Intake			

If the valve is **not** within specification, it **must** be replaced.

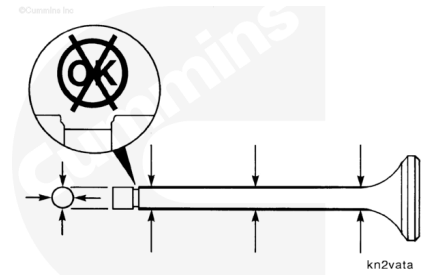
It is **not** necessary to remove any black or brown stains on the valve closest to the port, unless the buildup affects the movement in the guide. Use nothing more than course crocus cloth (1000 grit) to remove the stains.

The valve stems are chrome plated. If there are scuffs or marks that can be felt with the fingernail, the valve **must** be replaced.

Check the collet groove area for wear. If the groove area is worn, the valve **must** be replaced.

Measure the valve stem outside diameter.

Valve Stem Outside Diameter		
mm		in
10.490	MIN	0.4130
10.510	MAX	0.4138



Cummins Inc. recommends using new valve springs when the cylinder head is reconditioned.

A bent or broken spring **must not** be used again.

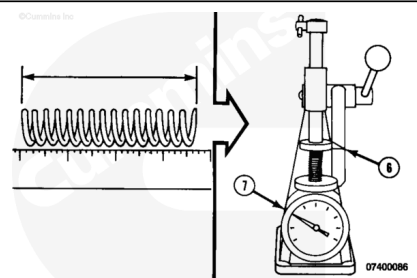
A spring with a notch worn in the first coil **must not** be used again. The ends of the springs will wear into the first coil, creating a worn notch in the material.

If a worn notch can be seen or felt, the valve spring **must not** be used again.

Often a spring that has these notches will make a clicking sound. To check for a click, the spring is to be hand compressed until the end of the first coil is completely collapsed. A large notch will cause a clicking sound to be heard.

Check both ends of the spring for the clicking sound.

Check the spring free length.



Approximate Free Length

87.63 mm [3.45 in]

Check the spring force at the indicated test height.

Valve Spring Test Height

57.23 mm [2.253 in]

Valve Spring Force

	n		lbf
	1477 N	MIN	332 lbf
	1637 N	MAX	368 lbf

If the valve spring is **not** within specification, the valve spring **must** be replaced.

with Electronically Actuated Injector

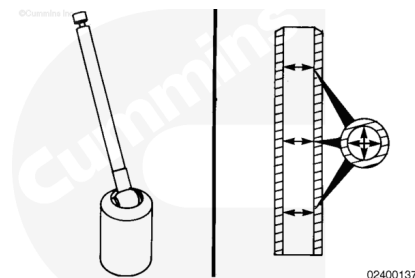
Check the valve guide for chips and cracks.

Measure the inside diameter of the guide with a ball gauge or a dial bore indicator.

Valve Guide Inside Diameter (Installed)

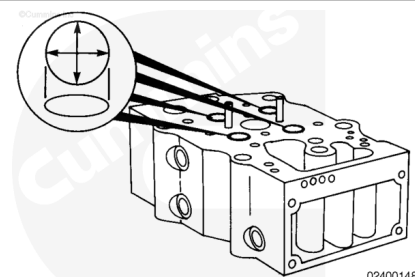
mm		in
12.600	MIN	0.496
12.667	MAX	0.499

The valve guide can be reusable if the inside diameter of the first 13 mm [0.50 in] from the top or bottom of the valve guide is over specification. However, if an area 13 mm [0.50 in] or greater from the top or bottom is out of tolerance, the valve guide **must** be replaced.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Check the valve guide for chips and cracks.

Clean the part with solvent, Part Number 3824421, or equivalent.

Measure the valve guide bore inside diameter.

Valve Guide Bore Inside Diameter		
mm		in
21.425	MIN	0.844
21.450	MAX	0.845

If the valve guide bore is **not** within specifications, ream the bore for an oversize valve guide.

The oversize valve guides are oversize on the outside diameter **only**. The bore in the valve guide is **not** oversize.

Valve guides are available in two oversizes:

- 0.25 mm [0.010 in]
- 0.38 mm [0.015 in].

Ream the valve guide bore to maintain a press fit within 0.028 to 0.067 mm [0.001 to 0.003 in] between the valve guide and the valve guide bore.

Clean the valve seat insert bore.

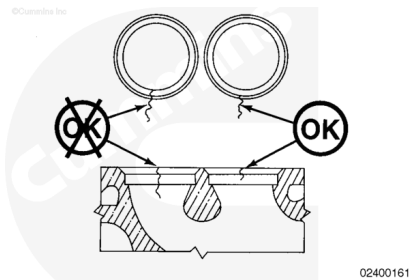
Check the length of any cracks extending into the valve insert bore.

The cylinder head **must** be replaced if a crack extends into the bottom of the bore.

Sometimes it is possible to remove the crack by machining the cylinder head to use an oversize valve seat insert.

Use valve insert counterbore cutter kit, Part Number ST-257, or equivalent, along with appropriate cutter listed below:

- ST-1310-1 (0.010 oversize)
- ST-1310-2 (0.020 oversize)
- ST-1310-3 (0.030 oversize)
- ST-1310-4 (0.040 oversize).



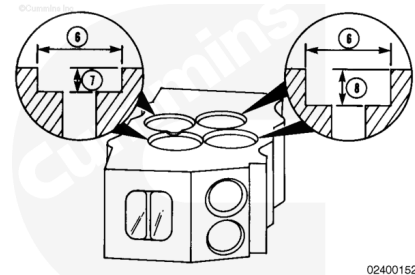
Measure the inside diameter and the depth of the valve seat insert bore.

Valve Seat Insert Bore Inside Diameter (6)
--

	mm		in
Exhaust and Intake	60.37	MIN	2.377
	60.40	MAX	2.378

Valve Seat Insert Bore Depth (7) and (8)

	mm		in
Exhaust and Intake	12.50	MIN	0.492
	12.62	MAX	0.497



02400152

If the valve guide seat insert inside diameter is **not** within specifications, use an oversize valve seat.

Oversize valve seat inserts are available in the sizes listed in the table below. Machine the valve seat insert bore in the cylinder head to maintain a press fit within 0.064 to 0.114 mm [0.002 to 0.005 in] between the valve seat insert and valve seat insert bore.

Outside Diameter Oversize	Depth (Thickness) of Oversize
0.25 mm [0.010 in]	Standard
0.51 mm [0.020 in]	0.13 mm [0.005 in]
0.76 mm [0.030 in]	0.25 mm [0.010 in]
1.02 mm [0.040 in]	0.38 mm [0.015 in]

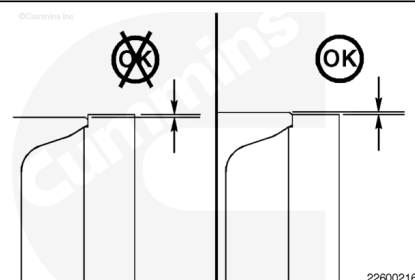
The intake and exhaust valve face and seating surfaces **must** be cleaned and free of carbon deposits before the inspection is performed.

Use Scotch-Brite™ 7448 abrasive hand pad, Part Number 3823258, or equivalent, to clean the valve.

The valve head thickness gauge, Part Number 3164983, is used to check intake and exhaust valve head thickness. The lower gauging surface marked "INTAKE" is for the intake valve. The upper gauging surface marked "EXHAUST" is for the exhaust valve.

Place the valve into the valve head thickness gauge.

Measure the height of the valve.



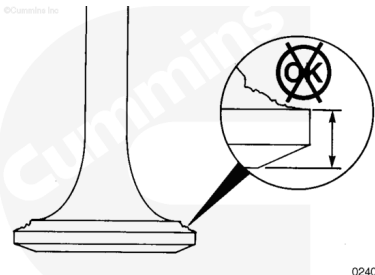
22600216

If the valve is flush or above the gauging surface, the valve can be used again.

If the valve is below the gauging surface, the valve can **not** be used.

⚠ CAUTION ⚠

This type of a check is not as accurate as the checking tool. It can result in valves that are not within minimum specification being reused. Valves that are not within minimum specification can fail, causing severe progressive damage to the power cylinder.



02400136

If a valve checking tool is **not** available, place the valve on a flat surface and check the height of the outside diameter.

Minimum Valve Head Thickness (at the outside diameter)

	mm		in
Exhaust	3.00	MIN	0.120
Intake	2.16	MIN	0.085

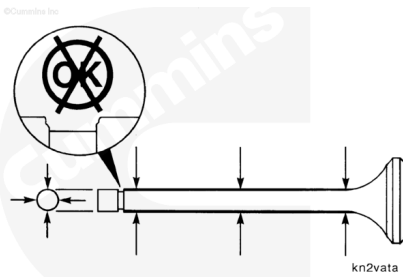
If the valve is **not** within specification, it **must** be replaced.

It is **not** necessary to remove any black or brown stains on the valve closest to the port, unless the buildup affects the movement in the guide. Use nothing more than course crocus cloth (1000 grit) to remove the stains.

The valve stems are chrome plated. If there are scuffs or marks that can be felt with the fingernail, the valve **must** be replaced.

Check the collet groove area for wear. If the groove area is worn, the valve **must** be replaced.

Measure the valve stem outside diameter.



kn2vata

Valve Stem Outside Diameter

mm		in
12.535	MIN	0.494
12.576	MAX	0.495

Valve springs of two different lengths are used on the engine. These are identified with a red strip (old spring) and a white strip (new spring)

Cummins Inc. recommends using white strip (new valve springs) when the cylinder head is reconditioned.

A bent or broken spring **must not** be used again.

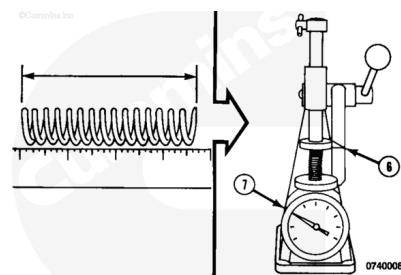
A spring with a notch worn in the first coil **must not** be used again. The ends of the springs will wear into the first coil, creating a worn notch in the material.

If a worn notch can be seen or felt, the valve spring **must not** be used again.

Often a spring that has these notches will make a clicking sound. To check for a click, the spring is to be hand compressed until the end of the first coil is completely collapsed. A large notch will cause a clicking sound to be heard.

Check both ends of the spring for the clicking sound.

Check the spring free length.



Approximate Free Length	
Red Strip Spring	White Strip Spring
65 mm [2.57 in]	69 mm [2.72 in]

Check the spring force at the indicated test height.

Test Height	
Red Strip Spring	White Strip Spring
39 mm [1.53 in]	38 mm [1.51 in]

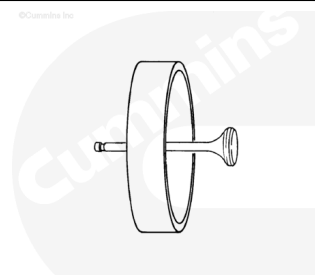
Valve Spring Force			
	n		lbf
Red Strip Spring	1183	MIN	266
	1308	MAX	294
White Strip Spring	1241	MIN	279
	1383	MAX	311

If the valve spring is **not** within specification, the valve spring **must** be replaced.

Magnetic Crack Inspect

⚠ CAUTION ⚠

To reduce the possibility of engine damage, always demagnetize and clean the parts thoroughly after a magnetic particle inspection.



kn2vakb

Use the magnetic particle residual method to check the valves for cracks.

Check the exhaust valves with the coil shot method.

Use a 305 mm [12 in] minimum diameter coil.

Coil Shot Amperage (Ampere Turns)

Minimum	Maximum
400-VDC or rectified VAC	800-VDC or rectified VAC

An ampere turn is a electrical current of one ampere flowing through the coil multiplied by the number of turns in the coil.

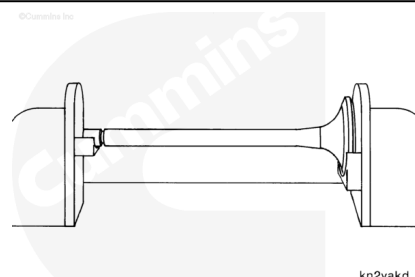
Test the valve.

A broad fuzzy pattern will appear at the welded joint on the exhaust valves. This is normal. If there is a distinct line in the broad fuzzy pattern, the valve **must** be replaced.

Use the head shot method to test the intake valve.

Head Shot Amperage (Ampere)

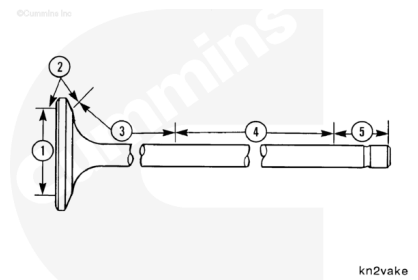
Minimum	Maximum
500-VDC or rectified VAC	700-VDC or rectified VAC



kn2vakd

The acceptable criteria for intake and exhaust data is listed below:

1. Indications less than 38.1 mm [1.50 in] in length are acceptable
2. No indications
3. **Only** longitudinal indications are acceptable
4. **Only** longitudinal indications are acceptable
5. No indications.

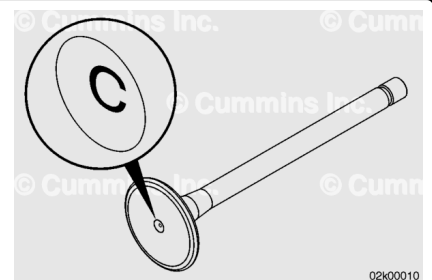


Grind

Valves

⚠ CAUTION ⚠

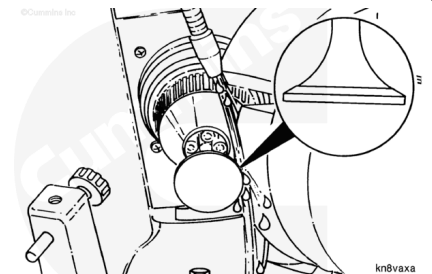
Valves marked with a "C" in the center of the valve head, must not be ground. Grinding these valves will damage the hardened layer of the valve face and severe engine damage can result.



Intake and exhaust valves **must** be ground to the same angle.

Use valve facing machine, Part Number 3376256, or equivalent, to grind the valve to a 30-degree angle, as shown in the illustration.

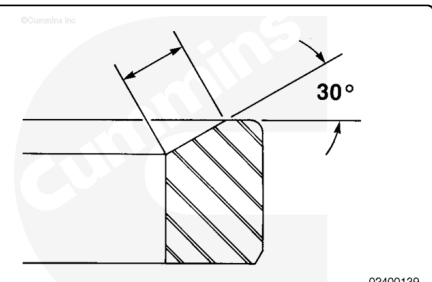
Make sure **not** to remove too much material too quickly. **Only** remove the minimum amount of material, making sure the seating area of the valve is free from grooves.



Machine

Valve Seat

Cummins Inc. recommends using machines that will cut the valve seat insert. Grinding the valve seat with stones that are too coarse can cause an unacceptable surface finish that can lead to early valve and/or seat wear. If the valve protrusion is too great and a machine to cut the valve seat is **not** available, Cummins Inc. recommends removing the seat, then machining the bore in the cylinder head to a greater depth, if possible.



Use a valve guide arbor set, Part Number 3375946, or equivalent, with valve seat grinding machining ST-685-A (110-VAC) or ST-685-C (220-VAC).

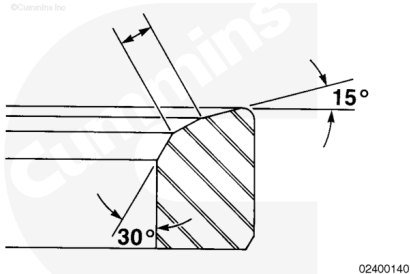
Machine the valve seat insert to the angle shown in the illustration.

Measure the valve seat insert width.

Valve Seat Insert Width			
	mm		in
Intake	3.05	MIN	0.120
	3.55	MAX	0.140
Exhaust	1.52	MIN	0.060
	2.54	MAX	0.100

If the width of the valve seat is **not** within specifications, remove the surface material on the inside diameter and outside diameter of the valve seating area to decrease the width of the valve seat.

If the valve seat specifications are **not** obtained by machining, the valve seat **must** be replaced.



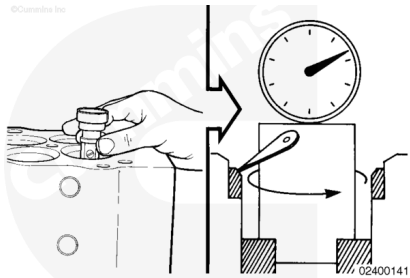
It is important that the eccentricmeter is **not** positioned too far down the arbor. The meter needle **must not** complete more than between one and one and a half revolutions before touching the valve seat insert. If the meter will **not** touch the insert at a minimum revolution, change the arbor to a smaller diameter. The meter will have to be adjusted for each seat and arbor combination.

Use an eccentricmeter, Part Number ST-685-4, or equivalent, and the arbor included in the valve seat grinding machine kit, Part Number ST-685-A, or equivalent, or Part Number ST-685-C, or equivalent.

Measure the valve seat to the valve guide concentricity.

The seat and guide **must** be concentric within 0.05 mm [0.002 in].

If the concentricity is **not** within specification, machine the valve seat.

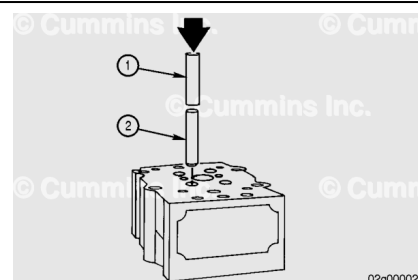


Assemble

with Mechanically Actuated Injector

Install the valve guides in all four locations in the cylinder head.

Use valve guide driver (1), Part Number 3163041, and an arbor press to install the valve guide (2) to the specified height

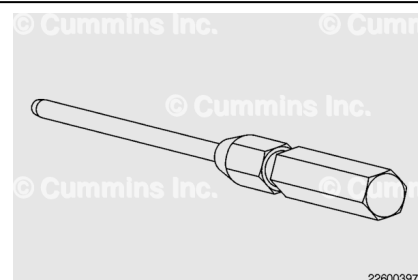


Valve Guide Height			
	mm		in
To spring pocket	37.97	MIN	1.495
	38.48	MAX	1.515
To top of head	13.97	MIN	0.550
	14.47	MAX	0.570

Inspect the valve guides for chips and cracks.

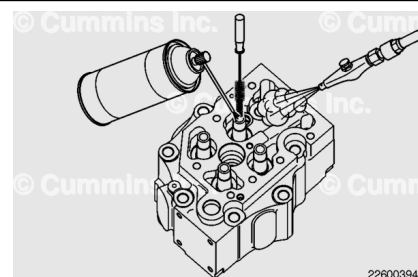
Use the drop gauge, Part Number 5298508, to check the valve guide for proper inside diameter and straightness.

Note : The drop gauge must be adjusted to a minimum of 114 mm [4.5 in] long.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

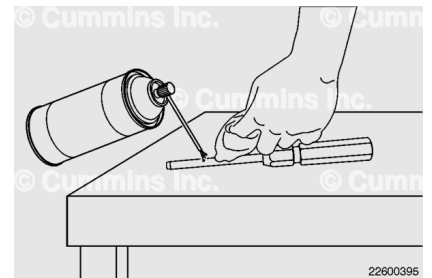
Clean the valve guide bores with an aerosol solvent (such as brake cleaner or carburetor cleaner) and a 7/16 inch diameter nylon brush.

Use compressed air to dry the valve guide bores.

Orient the cylinder head with the combustion face down, sitting flat on a level surface.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Clean the drop gauge pin. Use the same type of solvent that is used on the valve guide bores.

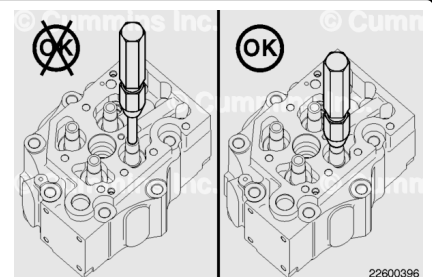
Note : Inspect the drop gauge pin for nicks, dings, scratches, and abrasion that can cause a false failure on the drop gauge test. Excessive wear can cause a false pass.

Holding the drop gauge by the upper end of the handle, place the tip of the gauge pin in the bore and let the gauge drop freely, by gravity alone.

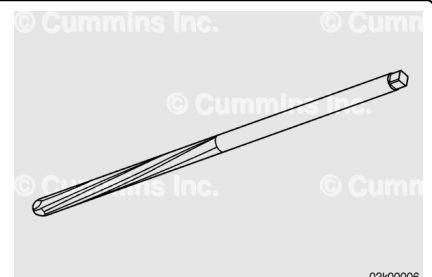
The valve guide passes if the gauge falls until the base of the handle contacts the valve guide.

If the drop gauge does **not** fall until the base of the handle contacts the valve guide, rotate the tool 90 degrees and test again.

If the drop gauge test fails at all four positions, the guide **must** be reamed and tested again.

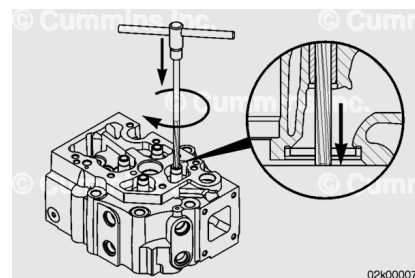


When a valve guide does **not** pass the drop gauge test, it **must** be reamed to bring the bore to the correct specification. The reamer, Part Number 5298510, is the **only** reamer to be used for this purpose. The reamer is to be used **only** by hand with a hand 'T' bar. It is **not** to be driven by any type of power drill or milling machine.

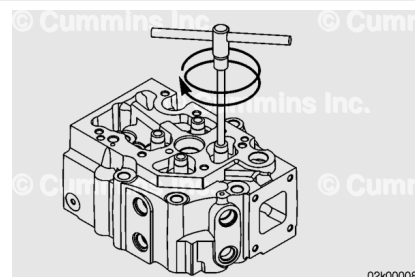


Note : It is essential that the reamer is continuously rotated while it is being inserted or removed from the valve guide. Damage to the valve guide will occur if the reamer is inserted or removed without rotation.

Position the cylinder head with the combustion face down. Insert the tip of the reamer into the top of the valve guide. While rotating the reamer **clockwise**, allow it to drop by gravity into the valve guide bore. Continue to rotate the reamer by hand, in a **clockwise** direction, until it drops through the valve guide and rests on the surface below.

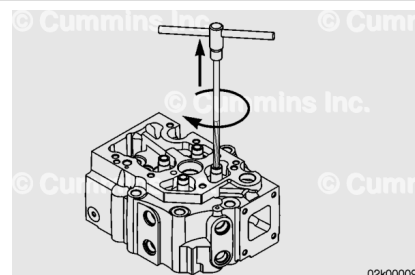


Once the reamer has dropped through the valve guide, rotate it through approximately two complete turns, continuing to rotate the reamer by hand, in a **clockwise** direction.

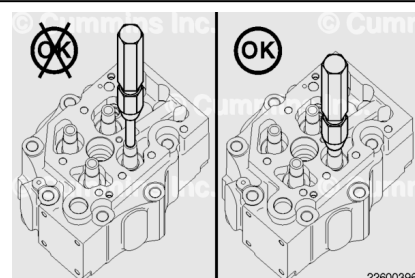


Note : It is essential that the reamer is continuously rotated while it is being inserted or removed from the valve guide. Damage to the valve guide will occur if the reamer is inserted or removed without rotation.

Withdraw the reamer while continuing to rotate it in a **clockwise** direction.



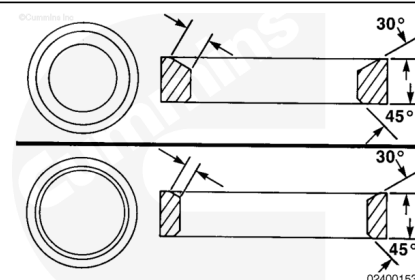
Make sure the valve guide is thoroughly cleaned. Repeat the drop gauge test as described above. If a valve guide does **not** pass the drop gauge test, repeat the reamer process until a drop gauge pass is achieved.



Identify the correct valve seat insert.

- The top illustration is the intake port valve seat insert with a wide valve seating area.
- The bottom illustration is the exhaust port valve seat insert.

Both the intake and exhaust port valve seat inserts **must** be installed with the 30-degree angle positioned as shown.

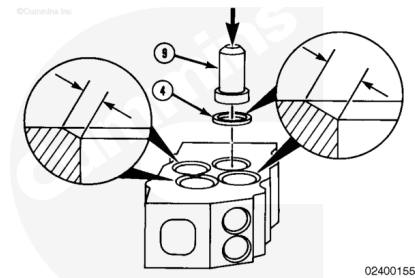


Exhaust valve seat inserts are color coded blue to identify the 30-degree angle side of the seat insert.

⚠ CAUTION ⚠

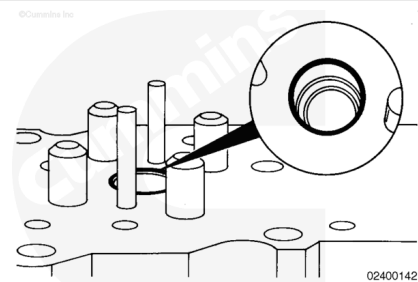
Do not use a hammer or mallet to install the valve seat inserts. The inserts can be cracked leading to a valve seat failure.

Install the valve seat insert (4) with a mandrel (9) and an arbor press.



Check the injector bore for cracks or other damage.

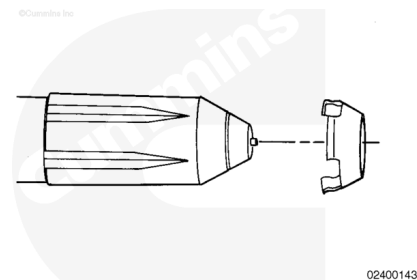
If cracks or other damage is found, the cylinder head **must** be repaired or replaced.



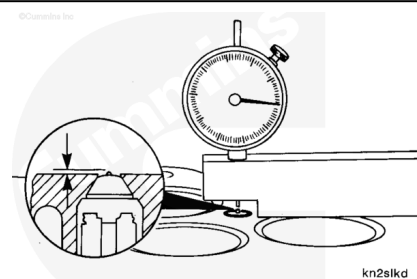
Install a sealing ring on a dummy injector.

Do **not** install the injector o-rings at this time.

Apply a thin coating of Prussian blue onto the sealing ring.



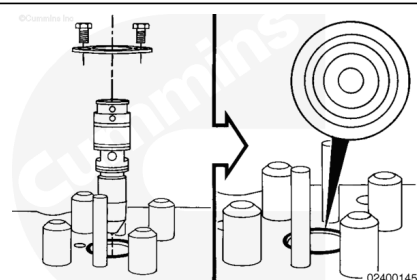
Measure the injector protrusion. Refer to Procedure 002-022 in Section 2. (</qs3/pubsys2/xml/en/procedures/20/20-002-022.html>)



Turn the cylinder head over to the original position.

Remove the injector hold-down clamp and the injector.

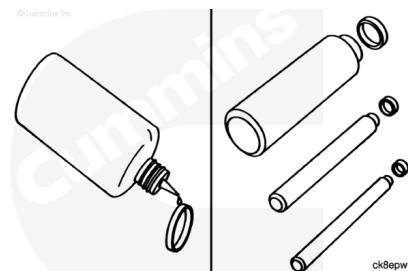
Check the contact pattern in the injector bore. A blue band a minimum of 1.52 mm [0.060 in] wide and 360 degree full circumference **must** be visible.



If the pattern does **not** meet specification, clean the bore of the injector with a tapered brush and check the seating pattern again.

The cylinder head **must** be repaired or replaced if the injector seating pattern is **not** within specifications.

Use cup plug Loctite™ sealant, Part Number 3375068.

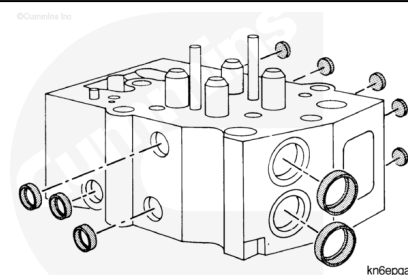


Use the correct expansion plug driver to make sure the expansion plugs are installed to the correct depth in the cylinder head. Drive the plug until the shoulder of the driver contacts the cylinder head.

- Part Number 3376813, expansion plug driver
- Part Number 3376814, expansion plug driver.

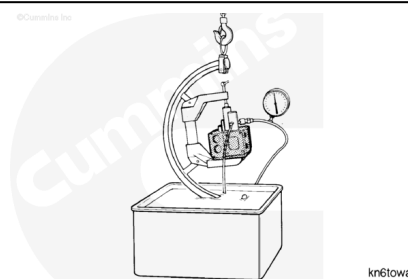
A cup plug driver handle, Part Number 3376795, or equivalent, **must** be used with expansion plug driver, Part Number 3376813, or equivalent, and expansion plug driver, Part Number 3376814, or equivalent.

Install ten new cup plugs.



Use a hydrostatic tester to check for evidence of leaks around cup plugs.

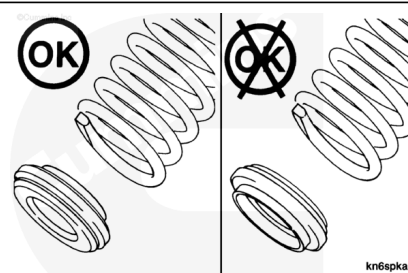
Reference the pressure test procedure.



When turning the rotator by hand, it can turn roughly, be difficult to turn, or **not** turn at all. This is normal until the rotator is installed. After installation, it will rotate freely.

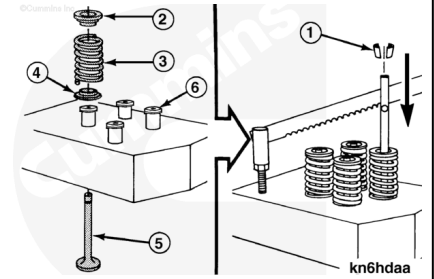
The rotator **must** be soaked in clean engine oil for at least 15 minutes prior to installation.

The rotator **must** be installed over the valve guide with the spring pilot flange face-up, as shown in this illustration.



⚠ WARNING ⚠

The valve springs are under compression. Use care when using the valve spring compressor. Personal injury can result if the tool slips from your hands.



Use a valve spring compressor, Part Number ST-448, Part Number 3375960, or equivalent. Valve spring stand, Part Number ST-1022, and compressor plate, Part Number 3375043, can be used to compress all four springs at the same time.

Install the parts.

1. Valve
2. Valve rotator
3. Valve spring
4. Valve spring retainer
5. Valve collet
6. Valve retainer.

Thoroughly lubricate the valve guide inside diameter with 140-weight gear oil before installing the valves.

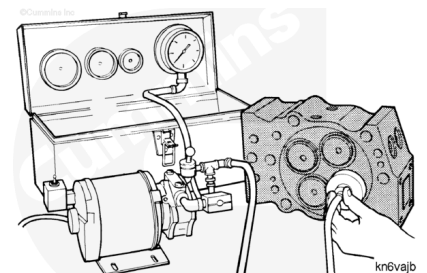
Use a valve vacuum tester, Part Number 3824277, or equivalent.

If the valve to the valve seat contact pattern appears to be correct and the valve still does **not** pass the vacuum test, perform the following check.

Apply a thin coating of grease on the outside diameter of the insert and the valve head. Repeat the vacuum test. The grease pattern will show the point of leakage.

If the leakage is between the insert and the valve, repair the valve seat.

If the leakage is between the insert and the head, the insert **must** be replaced.

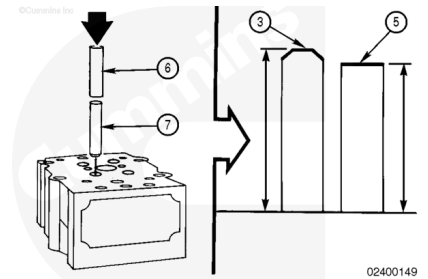


with Electronically Actuated Injector

Use the valve guide installation tool (6), Part Number 4918258, or equivalent, and an arbor press to install the valve guide (7) to the specified height.

Valve Guide Height (Installed)

mm		in
27.737	MIN	1.092
28.245	MAX	1.112



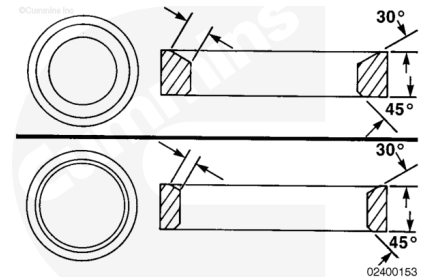
02400149

Identify the correct valve seat insert.

- The top illustration is the intake port valve seat insert with a wide valve seating area.
- The bottom illustration is the exhaust port valve seat insert.

Both the intake and exhaust port valve seat inserts **must** be installed with the 30-degree angle positioned as illustrated with the 45-degree angle to the cylinder head port.

Exhaust valve seat inserts are color coded blue to identify the 30-degree angle side of the seat insert.

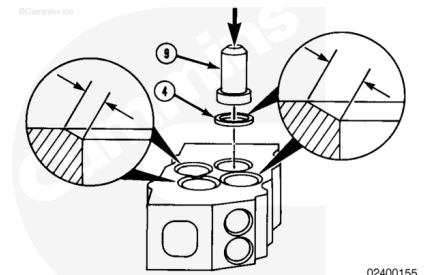


02400153

⚠ CAUTION ⚠

Do not use a hammer or mallet to install the valve seat inserts. The inserts can be cracked leading to the seat failure.

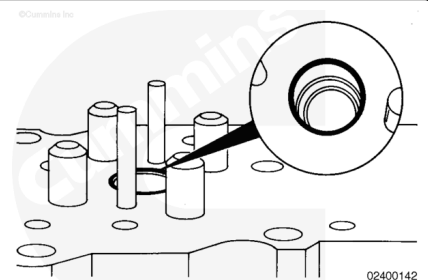
Install the valve seat insert (4) with a mandrel (9) and an arbor press.



02400155

Check the injector bore for cracks or other damage.

If cracks or other damage is found, the cylinder head **must** be repaired or replaced.

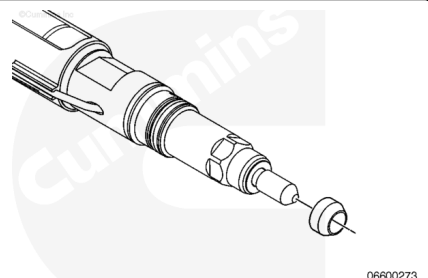


02400142

⚠ CAUTION ⚠

Support the cylinder head to prevent damage to the injector tip that protrudes from the combustion face.

Install a sealing ring on a dummy injector.



06600273

Do **not** install the injector o-rings.

Apply a thin coating of Prussian blue into the sealing ring.

Install the injector into the cylinder head. Refer to Procedure 006-026 in Section 6.

(/qs3/pubsys2/xml/en/procedures/20/20-006-026-tr.html)

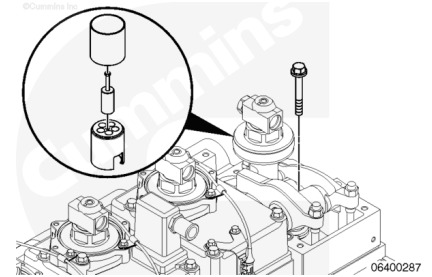
Note : There is **not** an injector protrusion check or a protrusion specification for electronically actuated injectors, as protrusion on electronically actuated injectors can **not** be adjusted.

Turn the cylinder head over to the original position.

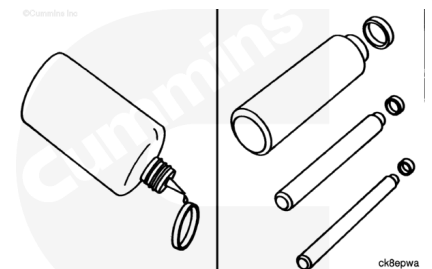
Remove the injector hold-down clamp and the injector.

Check the contact pattern in the injector bore. A blue band, a minimum of 1.52 mm [0.060 in] wide and 360 degrees full circumference **must** be visible. If the pattern does **not** meet specifications, clean the bore of the injector with a tapered brush and check the sealing pattern.

The cylinder head **must** be repaired or replaced if the injector pattern is **not** within specifications.

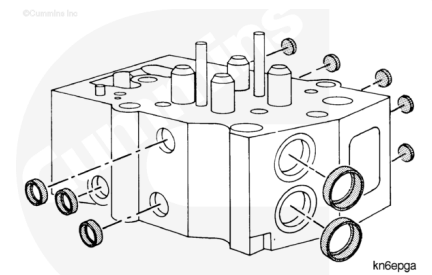


Apply Loctite™ 609 sealant, Part Number 3823718, or equivalent, to the cup plugs.



Cup plug driver handle, Part Number 3164085 or equivalent, is used with expansion plug drivers, Part Number 33376813, or equivalent, and Part Number 3376814, or equivalent.

Use the appropriate expansion plug driver and handle combination to drive the cup plug in until the shoulder of the driver contacts the cylinder head.

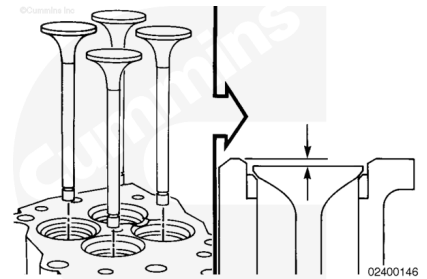


Install the valve into the valve guide.

Measure the depth of the valve with a depth gauge, Part Number 3164438, or equivalent.

The valve head **must** be even with, or **not** more than 0.51 mm [0.020 in] below, the surface of the cylinder head.

If the valve depth is **not** within specifications, the valve or the valve seat **must** be replaced.

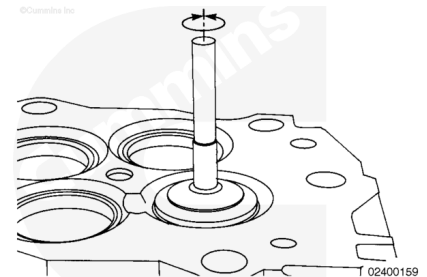


Apply a thin, even coating of fine lapping compound, Part Number 3375805, or equivalent, onto the valve face.

Provide pressure in the center of the valve with a power or hand suction lapping tool.

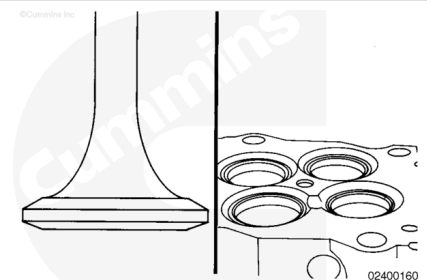
Turn the valve backward and forward.

Continue lapping until the compound shows a continuous contact pattern on both the valve seat insert and the valve.



⚠ CAUTION ⚠

Lapping compound is an abrasive material. Failure will result if the cylinder head, the valves, and the valve seats are not cleaned thoroughly.



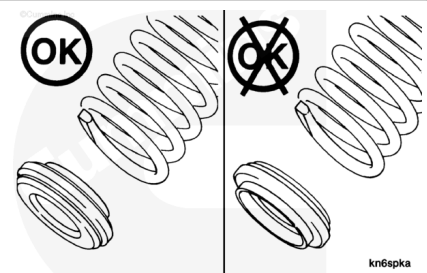
Remove the valve.

Clean the lapping compound from the valve, valve seat insert, and cylinder head.

When turning the rotator by hand, it can turn roughly, be difficult to turn, or **not** turn at all. This is normal until the rotator is installed; after installation, it will rotate freely.

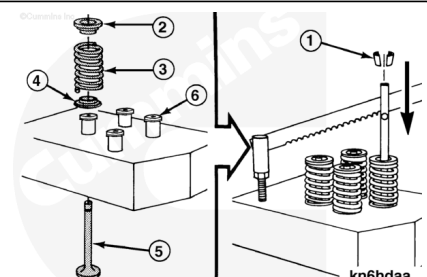
The rotator **must** be soaked in clean engine oil for at least 15 minutes prior to installation.

The rotator **must** be installed over the valve guide with the spring pilot flange face up, as shown in the illustration.



⚠ WARNING ⚠

The valve springs are under compression. Use care when using the valve spring compression tool. Personal injury can result if the tool slips out of the hands.



Use one of the below listed valve spring compressors to install the valve springs:

- Valve spring compressor, Part Number 3163606, or equivalent
- Valve spring compressor stand, Part Number ST-1022, or equivalent
- Valve spring compressor plate, Part Number 3163177, can be used with valve spring compressor, Part Number 3163606, or equivalent, and valve spring compressor stand, Part Number ST-1022, to remove four springs at a time
- Air operated valve spring compressor, Part Number 3375960.

Thoroughly lubricate the valve guide inside diameter with 140-weight gear oil.

Install the parts:

1. Valve
2. Valve rotator
3. Valve spring
4. Valve spring retainer
5. Valve collet
6. Valve retainer.

Note : Service tool, Part Number 4918258, or equivalent, is used to install both the valve guide to the proper height and install the valve stem seals as well.

Install the valve stem using valve guide installation tool, Part Number 4918258, or equivalent.

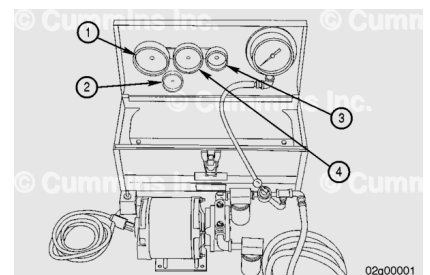
Vacuum Test

The valve vacuum tester, Part Number 3824277 (115-VAC, 50/60 hz) or 3824278 (220-VAC, 50/60 hz), can be used to test all Cummins® engine models.

Before using the tester, test the leakage shutoff valve. If the valve is dirty or worn, it will produce a false leakage measurement on the gauge.

Check the valve:

- Open the shutoff valve
- Turn on the vacuum pump



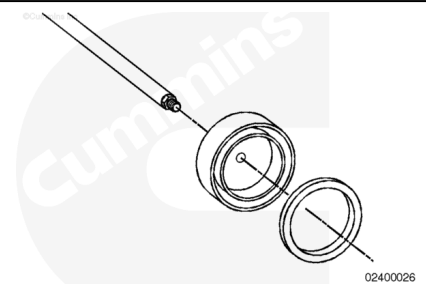
- Place the cup against a smooth surface
- Close the shutoff valve
- Turn off the vacuum pump
- Wait approximately 10 seconds
- The gauge **must not** drop more than 7 kPa [2 in Hg].

The valve and the valve seats **must** be clean and dry.

Choose the correct cup and the correct seal from the service tool kit for the engine model that is to be tested.

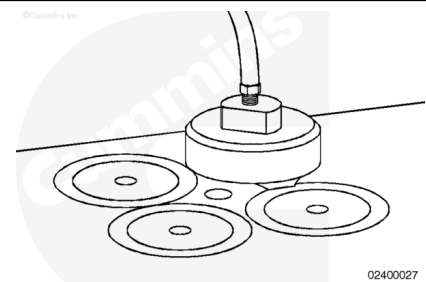
1. K19, K38, K5, QSK45, and QSK60
2. L10
3. 401cl and 464 cl
4. NH, V903, and V28

Install the seal and the cup to the vacuum line (hose).



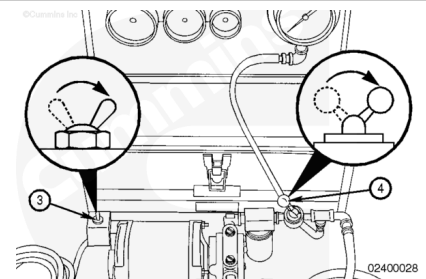
Cover the valve with the cup and the seal. The seal **must** have a tight contact on the cylinder head around the valve.

To check the exhaust valves, the seal **must** completely fill the milled area between the exhaust valves.



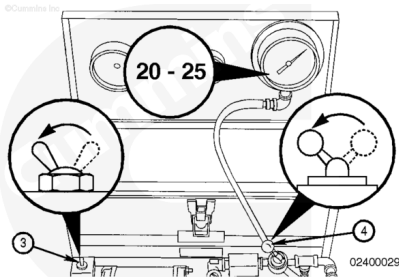
Move the toggle switch (3) to the ON position.

Turn the vacuum control valve (4) to the OPEN position.



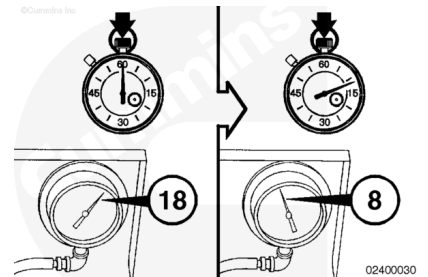
When the gauge indicates between 34 to 85 kPa [10 to 25 in-Hg], turn the vacuum control valve (4) to the CLOSED or OFF position.

Turn the toggle switch (3) to the OFF position.



⚠ CAUTION ⚠

The cylinder head must be disassembled and cleaned after any grinding or cutting procedures to reduce the possibility of engine damage.



Use a stopwatch. As the vacuum gauge needle moves **counterclockwise**, start timing when the needle on the gauge points to 61 kPa [18 in Hg].

Stop timing when the needle on the gauge points to 27 kPa [8 in Hg].

The elapsed time for the gauge to move between the specified readings **must** be 10 seconds or more.

If the elapsed time is less than 10 seconds, perform the following checks:

- Repeat the test to be certain the equipment is functioning properly.
- Use a mallet to lightly hit the valve stem to be certain the valve is sealed. Repeat the test.
- Apply a thin coating of grease on the outside diameter of the insert and the valve head. Repeat the vacuum test. The grease pattern will show the point of leakage.
- If the leakage is between the valve insert and the head, the insert **must** be replaced.

If the leakage is between the valve and the valve insert seat, one of the following procedures **must** be performed:

- Lap valve to insert seat
- Grind the valves
- Grind the valve insert seat.

Install

⚠ WARNING ⚠

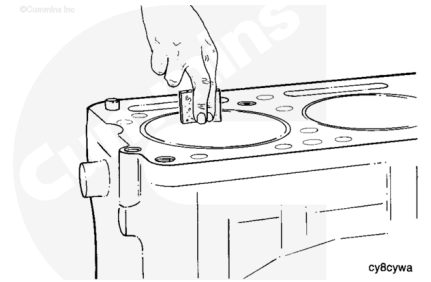
Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

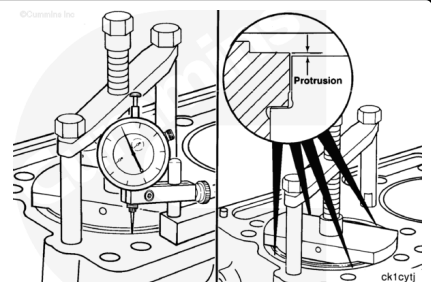


Clean the top of the cylinder block and the cylinder liners.

Clean the threaded holes which attach the cylinder head to the block with electrical contact cleaner, Part Number 3814510, or equivalent.

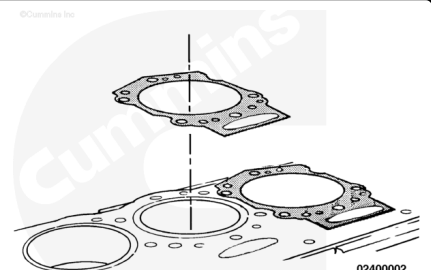
Dry with compressed air.

Measure the liner protrusion. Refer to Procedure 001-028 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-028-tr.html>)



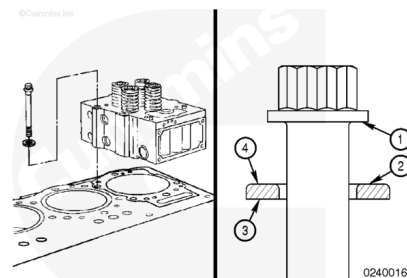
The word TOP, stamped on top of the head gasket, **must** be visible after the gasket is installed.

Install the cylinder head gasket.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Install the cylinder head. It **must** slide easily over the dowel pins.

Apply clean engine oil to the threads of the cylinder head capscrews.

Apply Loctite™ 51606, or equivalent, non-metal, graphite base, anti-seize compound to the top side of the washer.

Install the capscrews.

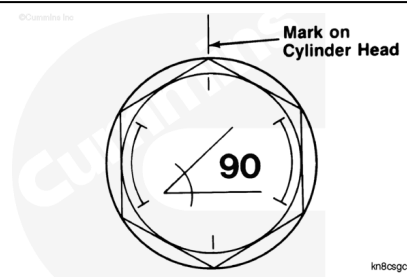
The markings on the head of the flange head capscrews serve as an aid during installation.

Torque Value:

Flange Head Capscrews

1. 298 n•m [220 ft-lb]

After tightening the capscrew, mark the cylinder head adjacent to one of the two single marks on the capscrew head.



Tighten the capscrews using the following steps in the sequence shown.

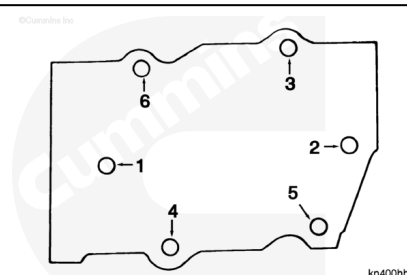
Torque Value:

1. 68 n•m [50 ft-lb]

2. 203 n•m [150 ft-lb]

3. 298 n•m [220 ft-lb]

4. Advance 90 degrees

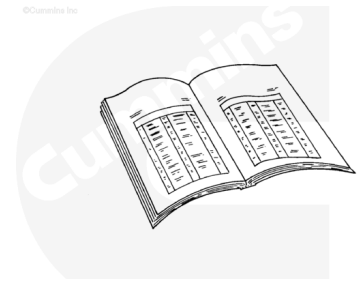


Finishing Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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- Install the injectors. Refer to Procedure 006-026 in Section 6. (</qs3/pubsys2/xml/en/procedures/20/20-006-026-tr.html>)
- Install the rocker lever housings, push rods, and crossheads. Refer to Procedure 003-013 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-013-tr.html>)
- Install the rocker levers. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-009-tr.html>)
- Install the high-pressure fuel lines to the injectors. Refer to Procedure 006-051 in Section 6. (</qs3/pubsys2/xml/en/procedures/20/20-006-051-tr.html>)
- Install the push rods or tubes. Refer to Procedure 004-014 in Section 4. (</qs3/pubsys2/xml/en/procedures/20/20-004-014-tr.html>)
- Adjust the valves. Refer to Procedure 003-006 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-006-tr.html>)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html>)
- Install the gear cover clamping plate. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-031-tr.html>)
- Install the coolant thermostat housing support. Refer to Procedure 008-015 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-015-tr.html>)
- Install the fuel tubes. Refer to Procedure 006-024 in Section 6. (</qs3/pubsys2/xml/en/procedures/20/20-006-024-tr.html>)
- Install the fuel manifolds. Refer to Procedure 006-022 in Section 6. (</qs3/pubsys2/xml/en/procedures/20/20-006-022-tr.html>)
- Install the aftercooler. Refer to Procedure 010-002 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-002-tr.html>)

010-002-tr.html)

- Install the exhaust manifold. Refer to Procedure 011-007 in Section 11. (/qs3/pubsys2/xml/en/procedures/20/20-011-007-tr.html)
- Install the turbocharger. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/20/20-010-033-tr.html)
- Connect the batteries. Refer to the OEM service manual.
- Connect the air starter supply line. Refer to the OEM service manual.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html)
- Operate the engine to 70°C [158°F] minimum coolant temperature and check for leaks.

Marine Applications

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

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⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift component or assembly.

⚠ CAUTION ⚠

Do not use gasket cement. It will prevent the gasket from sealing properly. Edge-molded gaskets do not require sealant.

⚠ CAUTION ⚠

Use caution when filling the cooling system so that the coolant is not spilled into the bilge area. Do not pump coolant overboard. If the coolant is spilled, it must be disposed of in accordance with local environmental regulations

- Install the injectors. Refer to Procedure 006-026 in Section 6. (</qs3/pubsys2/xml/en/procedures/20/20-006-026-tr.html>)
- Install the rocker lever housing. Refer to Procedure 003-013 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-013-tr.html>)
- Install the push rods and crossheads. Refer to Procedure 004-014 in Section 4. (</qs3/pubsys2/xml/en/procedures/20/20-004-014-tr.html>)
- Install the rocker levers. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-009-tr.html>)
- Adjust the valves and injectors. Refer to Procedure 003-006 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-006-tr.html>)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html>)
- Install the aftercooler assembly. Refer to Procedure 010-002 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-002-tr.html>)
- Install the exhaust manifold (dry). Refer to Procedure 011-007 in Section 11. (</qs3/pubsys2/xml/en/procedures/20/20-011-007-tr.html>)
- Install the exhaust manifold (wet). Refer to Procedure 011-008 in Section 11.

(/qs3/pubsys2/xml/en/procedures/20/20-011-008-tr.html)

- Install the turbocharger. Refer to Procedure 010-037 in Section 10. (/qs3/pubsys2/xml/en/procedures/20/20-010-037-tr.html)
- Install the exhaust manifold heat shields. Refer to Procedure 011-015 in Section 11. (/qs3/pubsys2/xml/en/procedures/20/20-011-015-tr.html)
- Install the turbocharger heat shields. Refer to Procedure 010-076 in Section 10. (/qs3/pubsys2/xml/en/procedures/20/20-010-076-tr.html)
- Install the aftercooler heat shields Refer to Procedure 010-129 in Section 10. (/qs3/pubsys2/xml/en/procedures/20/20-010-129.html)
- Connect the batteries. Refer to the OEM service manual.
- Fill the cooling system(s). Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html)
- Operate the engine to 70°C [158°F] minimum coolant temperature and check for leaks.

with Electronically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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- Install the injectors. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/20/20-006-026-tr.html)
- Install the rocker lever housings. Refer to Procedure 003-013 in Section 3. (/qs3/pubsys2/xml/en/procedures/20/20-003-013-tr.html)
- Install the gear cover clamping plate. Refer to Procedure 001-031 in Section 1. (/qs3/pubsys2/xml/en/procedures/20/20-001-031-tr.html)

- Install the thermostat housing. Refer to Procedure 008-015 in Section 8.
(/qs3/pubsys2/xml/en/procedures/20/20-008-015-tr.html)
- Install the push rods or tubes. Refer to Procedure 004-014 in Section 4. (/qs3/pubsys2/xml/en/procedures/20/20-004-014-tr.html)
- Install the rocker levers. Refer to Procedure 003-009 in Section 3. (/qs3/pubsys2/xml/en/procedures/20/20-003-009-tr.html)
- Adjust the valves and injectors. Refer to Procedure 003-006 in Section 3.
(/qs3/pubsys2/xml/en/procedures/20/20-003-006-tr.html)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html)
- Install the coolant thermostat housing. Refer to Procedure 008-015 in Section 8.
(/qs3/pubsys2/xml/en/procedures/20/20-008-015-tr.html)
- Install the intake manifold. Refer to Procedure 010-023 in Section 10. (/qs3/pubsys2/xml/en/procedures/20/20-010-023-tr.html)
- Install the fuel drain lines. Refer to Procedure 006-013 in Section 6. (/qs3/pubsys2/xml/en/procedures/20/20-006-013-tr.html)
- Install the high pressure fuel lines. Refer to Procedure 006-051 in Section 6.
(/qs3/pubsys2/xml/en/procedures/20/20-006-051-tr.html)
- Install the intake manifold. Refer to Procedure 010-023 in Section 10. (/qs3/pubsys2/xml/en/procedures/20/20-010-023-tr.html)
- Install the exhaust manifold. Refer to Procedure 011-007 in Section 11. (/qs3/pubsys2/xml/en/procedures/20/20-011-007-tr.html)
- Install the turbocharger. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/20/20-010-033-tr.html)
- Connect the batteries or air starter supply line. Refer to the OEM service manual.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html)

- Operate the engine to 70°C [158°F] minimum coolant temperature and check for leaks.

Last Modified: 03-Jun-2015
